

Carved-Path Pattern List — a thinking tool

*A reference object, not a finding. This is the full raw output of the carving analogy used as a pattern generator — laid out so it can be examined and thought against. It is **not** a taxonomy, **not** evidence of model internals, and **not** a protocol category list. Its companion framing — including why the high match-rate is a caution rather than a credential — is in open-question-uneven-support.md. Read that first; this list only makes honest sense alongside it.*

How to read this list (the one thing that matters)

Every row below “matches” a real LLM error. That is exactly why the list must be read carefully: **the matching is post-hoc**. Each pattern was named to fit an error already known, not predicted in advance — and every match reduces to a failure family the field established without any analogy. So the list is not a scorecard showing the analogy is right (a frame that fits everything proves nothing). It is a record of the analogy’s *generative reach*, kept as raw material for thinking, with the understanding that **reach is not validity**.

The two right-hand columns are deliberately not “does this error exist / how strong is the match” (the original framing, which dresses post-hoc fit as confirmation). They are: the **field-established family** the pattern collapses into, and its **status** — which is the same for almost every row, on purpose.

The list

#	Carved-path pattern	Behavioral error it suggests	Field-established family it reduces to	Status
1	Shallow groove	Vague, generic answer; loses specificity under stress	Stress-retention loss	post-hoc; field-owned
2	Incomplete groove	Starts correctly, fails before completing	First-error-step / truncated execution	post-hoc; field-owned
3	Discontinuous groove	Does A and B alone, fails A→B chain	Compositional-generalization failure	post-hoc; field-owned, possible test target
4	Mis-carved groove	Consistently follows a wrong heuristic	Shortcut learning	post-hoc; field-owned, correctness-

#	Carved-path pattern	Behavioral error it suggests	Field-established family it reduces to	Status
5	Stray groove	Veers off-topic / irrelevant association	Perturbation / attention drift	guard required post-hoc; field-owned
6	Over-deep groove	Overuses a memorized pattern inappropriately	Memorization / over-application	post-hoc; field-owned
7	Crossed grooves	Context conflict causes answer flip	Context contamination	post-hoc; field-owned
8	Braided grooves	Two patterns interfere; output blends them	Interference / fact mixing	post-hoc; field-owned
9	Eroded groove	Weakens under quantization, pruning, noise, long context	Stress-retention loss	post-hoc; field-owned
10	Narrow ridge / tight channel	Exact symbolic task fails from tiny perturbation	Perturbation sensitivity	post-hoc; field-owned
11	Wide basin	Fluency survives while precision decays	Fluency–correctness dissociation	post-hoc; field-owned
12	Forked path	Unstable answers across paraphrases	Prompt / paraphrase sensitivity	post-hoc; field-owned
13	False outlet	Coherent reasoning, wrong final answer	Fluency–correctness dissociation	post-hoc; field-owned
14	Dead end	Repeats, stalls, refuses, cannot continue	Degeneration / refusal	post-hoc; field-owned (looser fit)
15	Loop in path	Repetition loops / circular reasoning	Degeneration	post-hoc; field-owned
16	Blocked channel	Safety/format	Refusal / format	post-hoc; field-

#	Carved-path pattern	Behavioral error it suggests	Field-established family it reduces to	Status
		constraint blocks a possible answer	constraint	owned (looser fit)
17	Weak bridge	Intermediate state not preserved across steps	Chain / state-transfer failure	post-hoc; field-owned
18	Over-smoothed channel	Collapses distinctions, treats cases as same	Over-generalization	post-hoc; field-owned
19	Rough walls	Small wording → large output variation	Perturbation sensitivity	post-hoc; field-owned
20	Sediment / debris	Irrelevant context contaminates answer	Context contamination	post-hoc; field-owned
21	Shortcut trench	Exploits benchmark artifact / position cue	Shortcut learning / benchmark artifact	post-hoc; field-owned
22	Flooded channel	High-probability language overwhelms rare correct answer	Frequency bias	post-hoc; field-owned
23	Dry channel	Rare task/domain answered poorly	Long-tail / coverage weakness	post-hoc; field-owned
24	Parallel grooves	Plausible completions compete; framing decides	Prompt sensitivity	post-hoc; field-owned
25	Collapsed wall	One error cascades into many	Autoregressive error cascade	post-hoc; field-owned (confound-controlled)
26	Uneven depth	Easy cases	Stress-retention	post-hoc; field-

#	Carved-path pattern	Behavioral error it suggests	Field-established family it reduces to	Status
		survive stress; hard cases collapse	loss	owned
27	Wrong merge point	Combines facts from different entities/events	Interference / fact mixing	post-hoc; field-owned
28	Path too short	Local reasoning works, long-horizon fails	Composition / long-horizon failure	post-hoc; field-owned
29	Overfit groove	Works on benchmark format, fails reformat	Overfitting to format	post-hoc; field-owned
30	Hidden sinkhole	Fluent, confident, subtly false	Fluency–correctness dissociation	post-hoc; field-owned

What the list collapses to

Thirty rows reduce to about seven field-established families. This is the actual content — the rest is vocabulary:

- **Stress-retention loss** — rows 1, 9, 26 (and the project’s one analytic survivor lives near here)
- **Composition / state-transfer failure** — rows 3, 17, 28
- **Shortcut / robust-wrong behavior** — rows 4, 21
- **Perturbation / prompt sensitivity** — rows 5, 10, 12, 19, 24
- **Interference / contamination** — rows 7, 8, 20, 27
- **Fluency–correctness dissociation** — rows 11, 13, 30, 22
- **Autoregressive error cascade** — rows 2, 25

None of these were discovered by the analogy. All were established by the field. The list shows the analogy can be *mapped onto* them, fluently — which is the generative reach worth thinking about, and the unfalsifiability worth being wary of, in equal measure.

Using this to think differently (the legitimate purpose)

This is the part that makes the object worth keeping. As a thinking tool — not a claim — the list invites a few genuinely open questions:

- Is there a *generative grammar* under the seven families — a smaller set of underlying failure modes (under-support, over-support, poor-connection, interference, stress) that the thirty surface patterns are combinations of?
- Does the same small grammar describe failure in *other* learned systems (habits, motor skills, roads, workflows)? — the cross-domain question parked in the open-question note.
- For any row: **what observation would make it false?** Most rows, as written, forbid nothing. The ones that could be sharpened into a forbidden outcome are the only ones with any future. (Row 3 / composition and row 4 / shortcut are the closest, which is why they are the two pressure-test survivors.)

The discipline, carried from everything else: a row earns its way out of “thinking tool” only by becoming a behavioral signature with a control and a falsification path — at which point the carving vocabulary is dropped and only the measurement remains.

A reference to think with, Apiana AI. Not a finding, not a taxonomy, not evidence. Companion to open-question-uneven-support.md.